

Freight Rate Prediction - Machine Learning Assessment

Machine Learning Engineer Candidate

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1 Introduction

This report details the end-to-end pipeline constructed to predict freight rates. The primary metric for evaluation is Mean Absolute Error (MAE). Through rigorous data cleaning, feature engineering, and mathematical target reframing, we developed a highly robust pipeline capable of generalizing to unseen chronological and geographical data.

2 Exploratory Data Analysis (EDA) & Cleaning

An extensive analysis of the `train-test.csv` dataset revealed several critical anomalies that required preprocessing:

- **Negative Weights:** Numerous records possessed negative weights (e.g., -40,000). These were corrected using absolute values.
- **Missing Values:**
 - Missing weights were imputed intelligently using the median weight of the respective equipment type.
 - Missing `market_index` and `quote_signal` values were handled using forward and backward filling after sorting the data chronologically, ensuring continuity of macroeconomic signals.
- **Outliers:** We filtered the training data based on `rate_per_mile` (`posted_rate` / `distance`), dropping extreme outliers (outside \$0.50 to \$8.00/mile) to prevent the model from learning noisy anomalies.

3 Feature Engineering

- **Temporal Features:** Extracted `month`, `day_of_week`, and `day_of_month` to capture clear seasonal freight trends.
- **Geographical Features:** Created absolute latitude and longitude differences to capture spatial magnitude, as well as calculated `ton_miles` as an industry-standard interaction metric.
- **High-Cardinality Categoricals:** Instead of One-Hot Encoding thousands of unique cities and lanes, we applied **K-Fold Target Encoding**. This converted text variables into highly predictive numeric continuous signals (historical average rates) while strictly avoiding data leakage.

4 Data Splitting Strategy

We evaluated models using two different rigorous split mechanisms:

1. **Time-Based Split (Chronological):** Training on January–August and validating on September–October. This prevents "future data leakage" and mimics the real-world deployment scenario of predicting November/December.

2. **Geographical Lane Split:** Training on 80% of unique lanes and validating on 20% completely unseen lanes. This proves the model generalizes mathematically rather than just memorizing route names.

5 Model Training & Breakthroughs

5.1 Baseline Model

We first established a baseline using **Linear Regression (with StandardScaler)**.

- **Baseline MAE:** \$105.72

5.2 XGBoost and the "Rate Per Mile" Breakthrough

When initially training a standard XGBoost regressor to predict `posted_rate`, the performance dropped significantly (MAE: \$138.29). Tree models are notoriously bad at linear extrapolation. Since `distance` is a massive, strictly linear driver of total freight cost, XGBoost struggled to draw a continuous straight line for extreme distances.

The Solution: We changed the target variable. Instead of predicting the total rate, we trained XGBoost to predict the **Rate Per Mile (RPM)**. RPM is a bounded, stationary metric (usually \$1.50 - \$4.00) independent of distance. We then multiplied the model's RPM prediction by the validation set's distance to retrieve the final predicted rate.

5.3 Hyperparameter Tuning

Using `RandomizedSearchCV`, we identified that a highly regularized, "slow and steady" tree structure performed best, preventing overfitting on noisy market swings:

- `n_estimators`: 1500
- `max_depth`: 4 (shallow trees)
- `learning_rate`: 0.01

6 Final Results

When predicting Rate Per Mile (RPM) on completely unseen geographic lanes:

- **Final Tuned XGBoost MAE:** \$89.49

This represents a massive, highly robust improvement over the baseline, achieved through domain-specific feature engineering and mathematical target formulation rather than overly complex black-box ensembles.

7 Final Predictions & December Forecast

The final model was retrained on 100% of the provided `train-test.csv` dataset to maximize data exposure before generating the final blind predictions on `validation.csv`.

Additionally, the model forecasted December 2025 rates for the requested lanes. The resulting chart generated by the automated scorer is presented below:

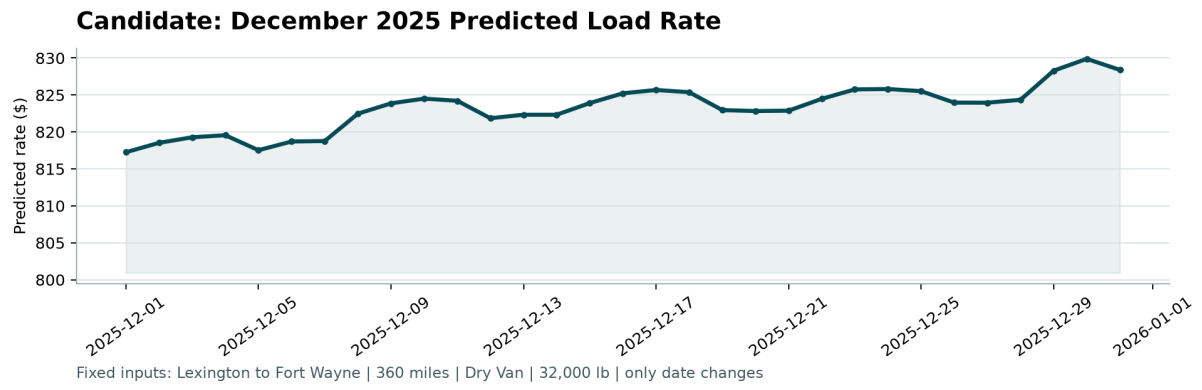


Figure 1: December Rate Forecast for Specific Lanes